Evidence that patients with rheumatoid arthritis have asymptomatic 'non-significant' Proteus mirabilis bacteriuria more frequently than healthy con...

OBJECTIVES: patients with rheumatoid arthritis (RA) are reported to have in their sera raised levels of antibody specific to Proteus mirabilis. The aim of the study was to verify this and to determine an explanation for it by investigating the frequency of P. mirabilis urinary tract infection in RA patients and matched controls. METHODS: freshly voided urine was examined for the presence, number and identity of infecting bacteria. The levels of antibody in blood and in urine of the IgM, IgA and IgG classes to the common O serotypes of P. mirabilis and the antigens to which they reacted were determined by enzyme-linked immunosorbent assay (ELISA) and immunoblotting. RESULTS: analysis of urine from 76 patients with RA and 48 age- and gender-matched healthy controls showed that only two (4%) of the control urines but 25 (33%) of those from the RA patients were infected. The commonest infecting organism in the RA patients' urine was Proteus mirabilis which occurred twice as frequently as Escherichia coli. Proteus mirabilis was found in 52% of the infected urines of the RA patients and was always detected as a pure growth and usually in insignificant (< 10(4)/ml) numbers. It is highly improbable that this finding was the outcome of differences in age, physical ability or medication between the RA and control patient groups. Comparison of antibody levels to P. mirabilis by ELISA showed RA patients had raised ($P < 0.0001$, $P = 0.001$, $P = 0.0063$) levels of IgA, IgG and IgM respectively in their sera and raised ($P < 0.0001$, $P < 0.0001$, $P = 0.0001$) levels of IgG, IgM and IgA respectively in their urine compared with the control group. It was not possible to detect an antibody reacting to a P. mirabilis antigen that was specific to the RA patients. CONCLUSION: the results confirm that RA patients have raised levels of antibody to P. mirabilis not only in blood but also in urine and suggest that this arises because RA patients have an asymptomatic, non-significant P. mirabilis bacteriuria more frequently or more prolonged than control patients. This may be the trigger for their RA condition.